

Lesson 30: Arc Length of Plane Curve and Parametric Curve

- **1. Formulation and Conceptual Understanding (20-30%)**
 - Correct identification of the arc length formula for the given curve type
(Cartesian: $L = \int \sqrt{1 + (f'(x))^2} dx$; Parametric: $L = \int \sqrt{(x'(t))^2 + (y'(t))^2} dt$).
 - Correct identification of the limits of integration ($a \leq x \leq b$ or $\alpha \leq t \leq \beta$).
 - Ensuring the curve is "smooth" (derivatives are continuous and the curve is traversed exactly once).
- **2. Differentiation and Substitution (20-30%)**
 - Accurate calculation of the derivatives $\frac{dx}{dt}$ and $\frac{dy}{dt}$ for parametric curves.
 - Correct substitution of the derivatives into the arc length formula.
 - Proper simplification of the expression inside the square root ($\sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2}$).
- **3. Integration Setup and Technique (20-30%)**
 - Setting up the definite integral with the correct bounds.
 - Applying appropriate integration techniques (e.g., u -substitution, trigonometric substitution, or numerical methods) to solve the integral.

- **4. Accuracy and Final Answer (10-20%)**
 - Correct evaluation of the definite integral.
 - Appropriate units and numerical approximation (if required).